

Model structure

MIPT

2026

Model selection

First level: select optimal parameters:

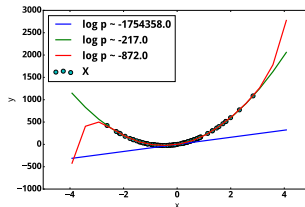
$$w = \arg \max \frac{p(\mathcal{D}|w)p(w|h)}{p(\mathcal{D}|h)},$$

Second level: select model optimizing Evidence:

$$p(\mathcal{D}|h) = \int_w p(\mathcal{D}|w)p(w|h)dw.$$



Model selection scheme

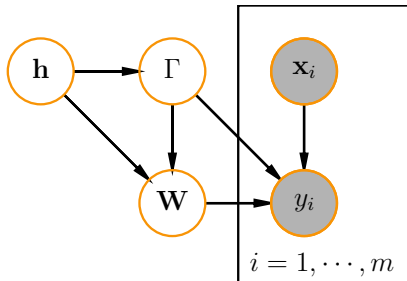


Example

Prior

Definition

Prior for parameters w and structure Γ of the model f is a distribution $p(W, \Gamma | h) : \mathbb{W} \times \Gamma \times \mathbb{H} \rightarrow \mathbb{R}^+$, where $\mathbb{W}$ is a parameter space, Γ is a structure space.



Definition

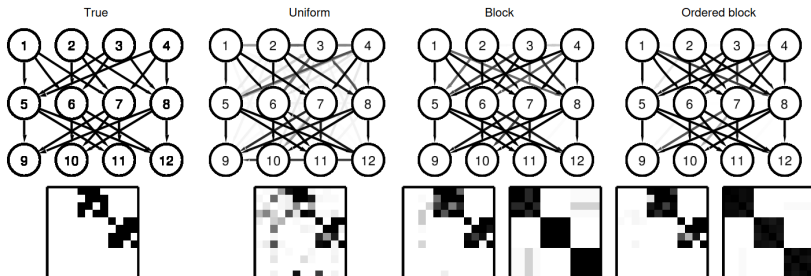
Hyperparameters $h \in \mathbb{H}$ of the models are the parameters of $p(w, \Gamma | h)$ (parameters of prior f).

Model f is assigned by the following values:

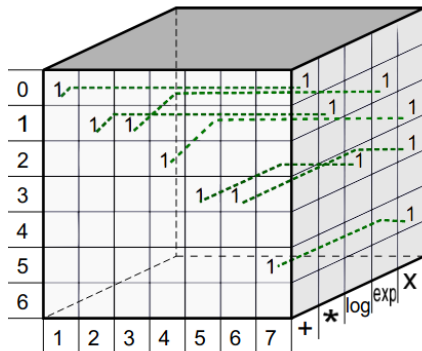
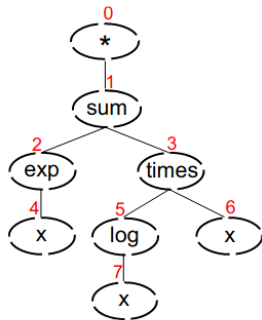
- **Parameters** $w \in \mathbb{W}$ set the superposition of submodels f_v .
- **Structure parameters** $\Gamma = \{\gamma^{j,k}\}_{(j,k) \in E} \in \Gamma$ set importance of each submodel f_v .
- **Hyperparameters** $h \in \mathbb{H}$ set prior distribution .
- **Metaparameters** $\lambda \in \Lambda$ set optimization function.

How can we assign model structure?

Example: Bayesian networks



Example: prediction of ranking functions



$$f = \exp(x) + (\log x)x$$

Трехиндексная матрица связей Z_f дерева Γ_f

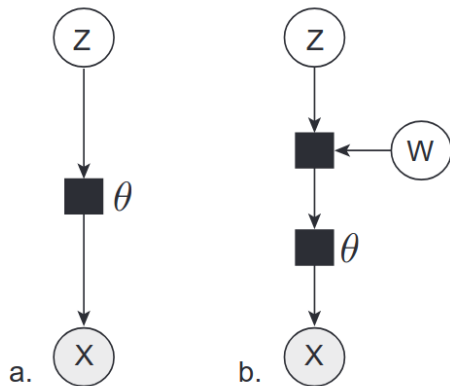
- вершины дерева пронумерованы;
- первые два индекса — номера вершин в ребре;
- третий индекс — выбранная элементарная функция на конце ребра.

Can we treat model structure as a parameter?

Example: Automatic relevance determination for VAE

$$p(x|w) = \int_{\gamma} \int_z p(x|z, \gamma, w) p(\gamma) p(z) d\gamma dz,$$

$$p(x|z, \gamma, w) = \text{MLP}(x, \gamma \cdot w).$$



Weight agnostic neural network

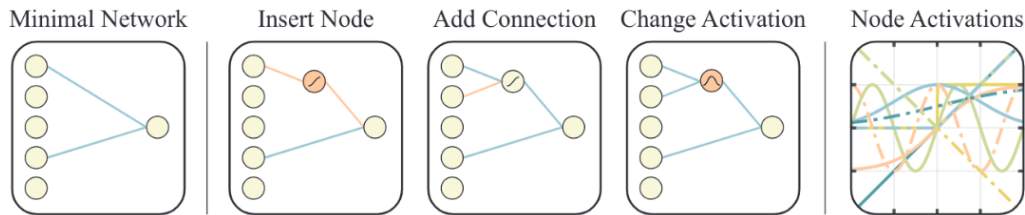


Figure 3: *Operators for Searching the Space of Network Topologies*

Left: A minimal network topology, with input and outputs only partially connected.

Middle: Networks are altered in one of three ways. *Insert Node:* a new node is inserted by splitting an existing connection. *Add Connection:* a new connection is added by connecting two previously unconnected nodes. *Change Activation:* the activation function of a hidden node is reassigned.

Right: Possible activation functions (linear, step, sin, cosine, Gaussian, tanh, sigmoid, absolute value, invert (i.e. negative linear), ReLU) shown over the range $[2, 2]$.

Feature selection

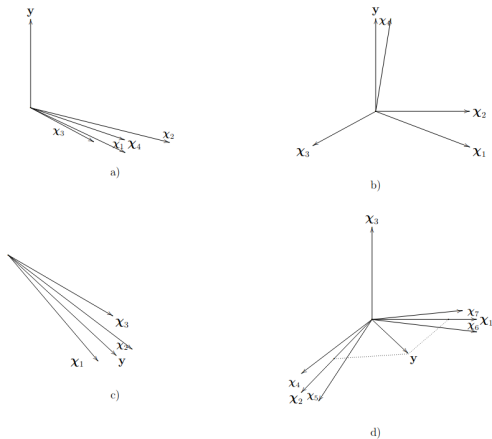


Figure 1: Synthetic test data sets configuration: **a)** inadequate and correlated, **b)** adequate and random, **c)** adequate and redundant, **d)** adequate and correlated.

Note: the problem becomes closer to the structure selection problem with relaxation methods development.

Optimal Brain Damage

The problem of removing irrelevant parameters (pruning) is considered.

Idea: Consider Taylor series for maximum point θ^* :

$$L(\theta^* + \Delta\theta) - L(\theta^*) = -\frac{1}{2}\theta^T H \theta + o(\|\Delta\theta\|^3),$$

where H is Hessian of $-L$.

Diagonalize the Hessian:

$$L(\theta^* + \Delta\theta) \rightarrow \max$$

where

$$\theta_i^* + \Delta\theta_i = 0.$$

Relevance of parameter:

$$\frac{\theta_i^2}{2[H^{-1}]_{i,i}}.$$

Importance estimation for neural network pruning

Basic idea is based on OBD-like Taylor expansion:

$$Q_i = (L(w) - L(w|w_i = 0))^2,$$

we can perform first order or second order approximation:

$$Q_i^1 = (\nabla_{w_i} L \cdot w_i)^2, \quad Q_i^2 = (\nabla_{w_i} L \cdot w_i - \frac{1}{2} w_i H_{i, \cdot} w)^2.$$

Importance estimation for neural network pruning

Group contribution:

$$Q^1 = (\sum_i \nabla_{w_i} L \cdot w_i)^2.$$

Alternatively we can assign importance of a set of parameters via gating:

$$w \cdot \gamma, \gamma \in \{0, 1\}, \quad Q^1 = \left(\frac{\partial L}{\partial \gamma} \right)^2 = (\sum_i \nabla_{w_i} L \cdot w_i)^2.$$

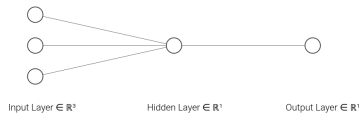
Note on the relation to information theory: Let $h = \frac{\partial L}{\partial \gamma}$. If the model and structure is converged ($E(h)^2 = 0$), then

$$\text{Var}(h) = E(h^2) - (Eh)^2 = Q^1.$$

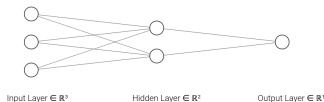
If L is a log-likelihood, then taking expectation (in variance) across population we get $\text{Var}(h)$ is a Fisher information.

Structure selection example

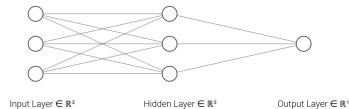
hidden layer dim = 1



hidden layer dim = 2



hidden layer dim = 3



All these models can be represented as $f(x, w) = \sigma \left((w^2)^T \sigma \left((w^1)^T x \right) \right)$
with similar shape of w^1 : $\dim(w^1) = 3 \times 3$.

Structure selection: one-layer network

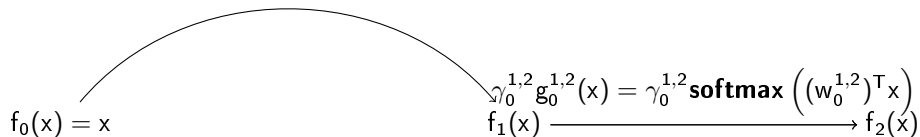
The model f is defined by the **structure** $\Gamma = [\gamma^{0,1}, \gamma^{1,2}]$.

$$\text{Model: } f(x) = \mathbf{softmax} \left((w_0^{1,2})^T f_1(x) \right), \quad f(x) : \mathbb{R}^n \rightarrow [0, 1]^{|Y|}, \quad x \in \mathbb{R}^n.$$

$$f_1(x) = \gamma_0^{0,1} g_0^{0,1}(x) + \gamma_1^{0,1} g_1^{0,1}(x),$$

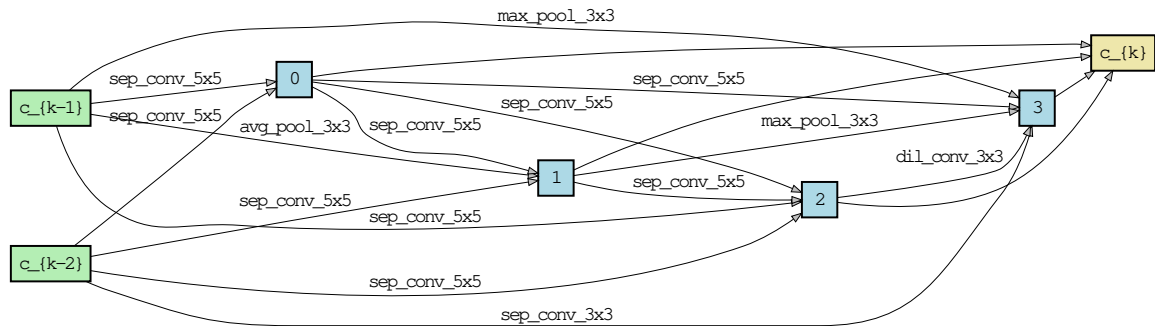
where $w = [w_0^{0,1}, w_1^{0,1}, w_0^{1,2}]^T$ — parameter matrices, $\{g_{0,1}^0, g_{0,1}^1, g_{1,2}^0\}$ — generalized-linear functions, alternatives of layers of the network.

$$\gamma_0^{0,1} g_0^{0,1}(x) = \gamma_0^{0,1} \sigma \left((w_0^{0,1})^T x \right)$$



$$\gamma_1^{0,1} g_1^{0,1}(x) = \gamma_1^{0,1} \sigma \left((w_1^{0,1})^T x \right)$$

Neural architecture search example



Structure selection: neural architecture search space

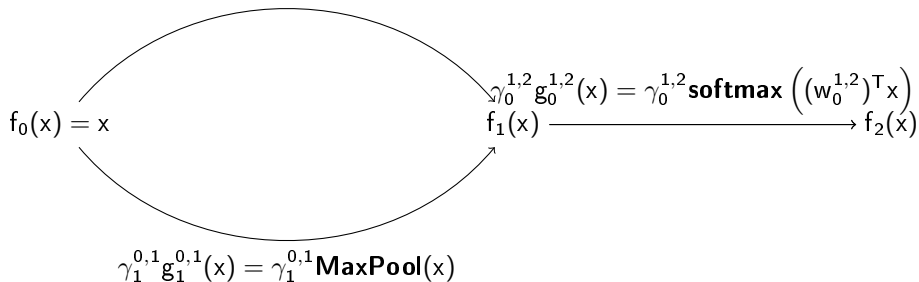
The model f is defined by the **structure** $\Gamma = [\gamma^{0,1}, \gamma^{1,2}]$.

$$\text{Model: } f(x) = \mathbf{softmax} \left((w_0^{1,2})^T f_1(x) \right), \quad f(x) : \mathbb{R}^n \rightarrow [0, 1]^{|Y|}, \quad x \in \mathbb{R}^n.$$

$$f_1(x) = \gamma_0^{0,1} g_0^{0,1}(x) + \gamma_1^{0,1} g_1^{0,1}(x),$$

where $w = [w_0^{0,1}, w_0^{1,2}]^T$ — parameter matrices, $g_{0,1}^0$ is a convolution, $g_{0,1}^1$ is a pooling operation, $g_{1,2}^0$ is a generalized-linear function.

$$\gamma_0^{0,1} g_0^{0,1}(x) = \gamma_0^{0,1} \mathbf{Conv}(x, w_0^{0,1})$$



Deep learning model structure as a graph

Define:

- ① acyclic graph (V, E) ;
- ② for each edge $(j, k) \in E$: a vector primitive differentiable functions $g^{j,k} = [g_0^{j,k}, \dots, g_{K^{j,k}}^{j,k}]$ with length of $K^{j,k}$;
- ③ for each vertex $v \in V$: a differentiable aggregation function $\mathbf{agg}_v$.
- ④ a function $f = f_{|V|-1}$:

$$f_v(w, x) = \mathbf{agg}_v \left(\{ \langle \gamma^{j,k}, g^{j,k} \rangle \circ f_j(x) \mid j \in \text{Adj}(v_k) \} \right), v \in \{1, \dots, |V| - 1\}, \quad f_0(x) = x \quad (1)$$

that is a function from $\mathbb{X}$ into a set of labels $\mathbb{Y}$ for any value of $\gamma^{j,k} \in [0, 1]^{K^{j,k}}$.

Definition

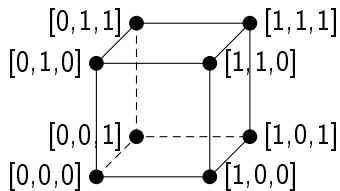
A *parametric set of models* $\mathfrak{F}$ is a graph (V, E) with a set of primitive functions $\{g^{j,k}, (j, k) \in E\}$ and aggregation functions $\{\mathbf{agg}_v, v \in V\}$.

Statement

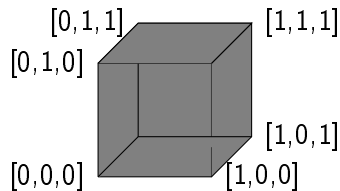
A function $f \in \mathfrak{F}$ is a model for each $\gamma^{j,k} \in [0, 1]^{K^{j,k}}$.

Structure restrictions

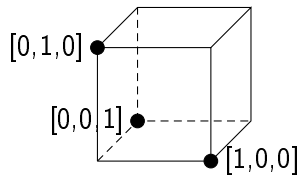
An example of restrictions for structure parameter γ , $|\gamma| = 3$.



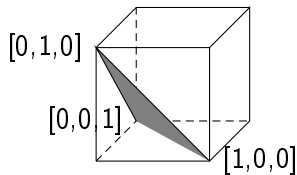
Cube vertices



Cube interior



Simplex vertices

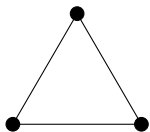


Simplex interior

Prior distribution for the model structure

Every point in a simplex defines a model.

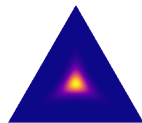
Gumbel-Softmax distribution: $\boldsymbol{\Gamma} \sim \text{GS}(s, \lambda_{\text{temp}})$



$$\lambda_{\text{temp}} \rightarrow 0$$

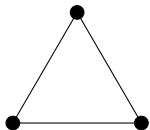


$$\lambda_{\text{temp}} = 0.995$$



$$\lambda_{\text{temp}} = 5.0$$

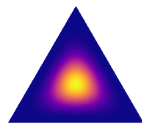
Dirichlet distribution: $\boldsymbol{\Gamma} \sim \text{Dir}(s, \lambda_{\text{temp}})$



$$\lambda_{\text{temp}} \rightarrow 0$$



$$\lambda_{\text{temp}} = 0.995$$



$$\lambda_{\text{temp}} = 5.0$$

Neural Architecture Search: problem statement

w are model parameters.

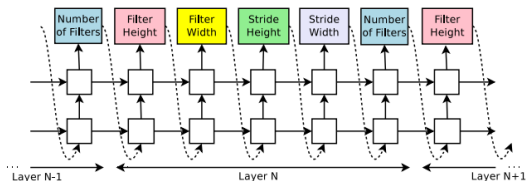
Γ is a structure.

$$\Gamma^* = \arg \max Q(w^*, \Gamma),$$

$$w^* = \arg \max L(w, \Gamma).$$

Neural Architecture Search with Reinforcement Learning

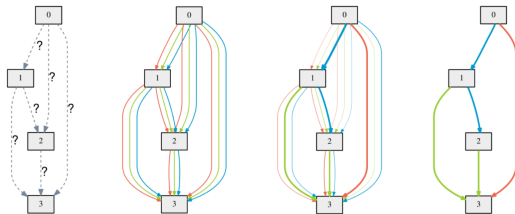
The structure is selected using controller.
The optimization of model parameters is conducted in a loop of structure selection.



DARTS

The model is a multigraph, where edges $[g^e]$ correspond to submodels, vertices $f_v(x)$ are the results of submodels:

$$f_v = \langle \gamma, \text{softmax}([g^e(x)]) \rangle.$$



DARTS

Optimization:

$$\Gamma^* = \arg \max Q(w^*, \Gamma),$$

$$w^* = \arg \max L(w, \Gamma).$$

The optimization is done using greedy gradient-like optimization:

$$\nabla_{\Gamma} Q(w', \Gamma) = \lambda_L \nabla_{\Gamma, w} L(w, \Gamma) \nabla_w Q(\Gamma, w').$$

Performance criteria for structure selection

- Number of parameters
- Number of vertices
- Number of edges
- Complexity of subfunctions

FBNet

$$\min_{\Gamma} \min_w L \cdot \lambda_1 \log \text{LAT}(\Gamma),$$

where LAT is a function of hardware latency of the operations for **target hardware**.

NAS with complexity control

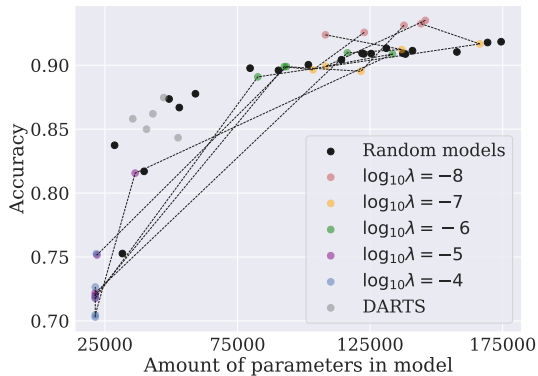
Our proposal

To use a mapping $\gamma(\lambda)$ instead of constant structural parameters $\gamma(\lambda)$, where λ is a regularization term for the loss function:

$$E_{\lambda} \left(\log p(y|X, w, \Gamma(\lambda)) + \lambda \sum_{(i,j)} \langle \text{softmax} \left(\gamma(\lambda)^{(i,j)} \right), n(g^{(i,j)}) \rangle \right),$$

where $n(g^{(i,j)})$ is a vector of amount of parameters for all the basic functions g .

Example: CIFAR-10



$$E_{\lambda} \left(\log p(y|X, w, \Gamma(\lambda)) + \lambda \sum_{(i,j)} \langle \text{softmax} \left(\gamma(\lambda)^{(i,j)} \right), n(g^{(i,j)}) \rangle \right).$$

Where it goes

Architecture	Source	Test Error		Params (M)	Search Cost			Comparable Search Space?	Search Method
		Best	Average		Stage 1	Stage 2	Total		
Shake-Shake [#]	[9]	N/A	2.56	26.2	-	-	-	-	manual
PyramidNet	[48]	2.31	N/A	26	-	-	-	-	manual
NASNet-A ^{**}	[52]	N/A	2.65	3.3	-	-	2000	N	RL
AmoebaNet-B [*]	[43]	N/A	2.55 $\pm$ 0.05	2.8	-	-	3150	N	evolution
ProxylessNAS [†]	[7]	2.08	N/A	5.7	4	N/A	N/A	N	gradient-based
GHN ^{#†}	[50]	N/A	2.84 $\pm$ 0.07	5.7	0.84	N/A	N/A	N	hypernetwork
SNAS [†]	[47]	N/A	2.85 $\pm$ 0.02	2.8	1.5	N/A	N/A	Y	gradient-based
ENAS [†]	[41]	2.89	N/A	4.6	0.5	N/A	N/A	Y	RL
ENAS	[34]	2.91	N/A	4.2	4	2	6	Y	RL
Random search baseline	[34]	N/A	3.29 $\pm$ 0.15	3.2	-	-	4	Y	random
DARTS (first order)	[34]	N/A	3.00 $\pm$ 0.14	3.3	1.5	1	2.5	Y	gradient-based
DARTS (second order)	[34]	N/A	2.76 $\pm$ 0.09	3.3	4	1	5	Y	gradient-based
DARTS (second order) [‡]	Ours	2.62	2.78 $\pm$ 0.12	3.3	4	6	10	Y	gradient-based
ASHA baseline	Ours	2.85	3.03 $\pm$ 0.13	2.2	-	-	9	Y	random
Random search WS [‡]	Ours	2.71	2.85 $\pm$ 0.08	4.3	2.7	6	9.7	Y	random

Results for CIFAR-10: Random search with weight sharing is competitive with SOTA

Current directions

- Purely applied works (NAS in medical segmentation)
- Multi-objective/multi-domain/multi-task settings
- Surrogates-based NAS + evolutionary algorithms

Reference

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Organizational issues

- Technical meeting: next week
- Format: each team shows the basic code (PoC). You should have an idea how this will be transformed into demo.
- The draft version of blog-post, tech. report and documentation will be checked in offline-mode.
- All the presented materials must be stored at the github
 - ▶ for blog-post and tech. report, you can put read-only link for the overleaf if you write the post here.
 - ▶ Time-limit: 15 minutes.